

Sem Project-I Literature Review

Semester III

“CUSTOMER CHURN PREDICTION SYSTEM FOR SIM USERS”

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Academic Year: 2026–27

Abstract

Customer churn – a subscriber discontinuing or switching away from a telecom operator – is one of the most significant sources of revenue loss for SIM-based service providers. This project report identifies a practical problem in which churn is frequently detected only after a customer has already left, because retention decisions are often based on generic plan-wise assumptions or reactive complaint handling rather than a continuous, data-driven assessment of individual subscriber behaviour. Such delayed detection limits the effectiveness of retention offers and increases customer-acquisition cost, since retaining an existing subscriber is generally less expensive than acquiring a new one.

The proposed Customer Churn Prediction System for SIM Users is intended to provide an early, data-backed churn-risk assessment through a simple, dashboard-based interface. The system accepts subscriber-level information – tenure, recharge amount, recharge frequency, call usage, internet/data usage, plan type, payment behaviour and customer complaints – and uses a trained classification model to categorise each customer into a Low, Medium or High churn-risk band. The interface is designed so that a telecom analyst or customer-retention team member can use it without requiring knowledge of machine learning.

The Stage-I work described in the project report covers problem identification, review of machine-learning approaches for churn prediction, dataset study, requirement analysis, comparative analysis with existing systems and technology-stack selection. The planned Stage-II work includes data preprocessing, exploratory data analysis, feature selection, training and evaluating classification models (Logistic Regression, Decision Tree and Random Forest as candidate models), connecting the selected model to a Flask backend, and building a dashboard for presenting risk levels and retention recommendations.

Existing research shows that machine learning, and ensemble methods in particular, can be applied effectively to structured telecom subscriber data to anticipate churn before it occurs. However, the literature also indicates challenges involving class imbalance between churners and non-churners, dataset and operator-specific generalisation, the trade-off between predictive accuracy and interpretability, and the practical adoption of predictions by retention teams. The reviewed studies therefore provide a technical foundation for the proposed system while also indicating the importance of validation and careful interpretation before churn predictions are used to drive retention actions.

Keywords: Artificial Intelligence, Machine Learning, Customer Churn Prediction, Telecommunications, SIM Users, Ensemble Learning, Feature Importance, Retention Strategy, Decision Support System

1 Literature Review

The following studies were selected because they address one or more components of the proposed Customer Churn Prediction System for SIM Users: churn classification techniques, ensemble learning, explainability, class-imbalance handling, big-data platforms and comparative model evaluation. The discussion follows the organisation and paragraph-based style of the supplied sample literature review, while the technical content is based on the project report and verified external publications.

[1] Customer Churn Prediction for Telecommunication Companies using Machine Learning and Ensemble Methods

The application of machine learning to telecom churn has expanded from single classifiers toward ensemble strategies that combine multiple models to obtain more stable predictions. This study investigates churn prediction using a telecom customer-churn dataset and evaluates Random Forest, LightGBM, XGBoost, Logistic Regression, Decision Trees and a custom Artificial Neural Network, together with an ensemble-averaging approach.

The study reports that individual boosted-tree models such as LightGBM and XGBoost, along with Logistic Regression, performed competitively, while the ensemble-averaging method achieved an accuracy of 0.79 and a recall of 0.72 on the test data – slightly lower than the best individual models in this particular experiment.

For the proposed SIM-churn system, the main relevance is the general workflow of comparing several classifiers side by side rather than committing to a single algorithm in advance. The project report follows a similar comparative approach, planning to train Logistic Regression, Decision Tree and Random Forest models and select the best-performing one using accuracy, precision, recall and F1-score.

The study is relevant because it shows that ensemble averaging does not automatically outperform well-tuned individual classifiers, which supports the project's plan to evaluate multiple candidate models empirically rather than assuming that a more complex ensemble is always superior. [1]

[2] Enhancing Customer Retention in the Telecom Industry with Machine Learning-Driven Churn Prediction

Customer retention strategy depends on identifying not just which customers are likely to churn, but which behavioural signals drive that risk. This study presents a broad investigation of

machine-learning-driven churn prediction aimed specifically at improving retention outcomes in the telecom industry.

The work reviews and applies a range of classification and ensemble-learning techniques to subscriber data, situating churn prediction within the wider goal of targeted, cost-effective retention campaigns rather than prediction accuracy alone.

The proposed project shares this retention-oriented framing: churn prediction is presented as an input to a retention workflow rather than an end in itself, and the system architecture explicitly separates the predicted risk level (Low/Medium/High) from the downstream retention action that a telecom analyst may choose to take.

This study supports the project's decision to treat churn-risk classification as a decision-support output rather than a final verdict, reinforcing the need for a dashboard that communicates risk levels in a way that is directly actionable for a retention team. [2]

[3] Explaining Customer Churn Prediction in the Telecom Industry Using Tabular Machine Learning Models

Telecom subscriber data is fundamentally tabular – structured records of tenure, usage, billing and plan attributes – which makes tabular machine-learning models a natural fit for churn prediction, while also raising the question of how predictions should be explained to non-technical stakeholders.

The study applies tabular machine-learning models to churn prediction and emphasises explainability, examining how model outputs can be interpreted in terms of the underlying subscriber features rather than treated as an opaque score.

This is directly relevant to the proposed system's use of feature importance and its stated goal of producing an understandable, not just accurate, risk classification. The project report's planned dashboard is intended to surface the leading contributing factors behind a prediction, echoing the explainability emphasis of this study.

The work illustrates that, for a decision-support tool used by non-technical retention staff, model interpretability is as important as raw predictive performance, which motivates the proposed project's preference for models such as Decision Tree and Random Forest that support relatively transparent feature-importance analysis. [3]

[4] An Adaptive Learning Approach for Customer Churn Prediction Using Evolutionary Computation and Naïve Bayes

Telecom churn datasets are frequently imbalanced, since the number of customers who churn in a given period is usually much smaller than the number who remain, which can bias a classifier toward the majority (non-churn) class. This study proposes an adaptive learning approach that combines evolutionary computation with a Naïve Bayes classifier to address this challenge.

The approach uses an evolutionary search process to adapt the learning procedure to the characteristics of the churn dataset, rather than applying a fixed classifier without adjustment for class imbalance.

The proposed SIM-churn project does not currently implement evolutionary optimisation, but the underlying concern – that a naively trained classifier may under-predict the minority churn class – is directly relevant to the project’s planned preprocessing stage, which is expected to include resampling or class-weighting strategies before model training.

This study is relevant because it demonstrates that churn-specific adaptations to standard algorithms can improve results beyond what an unmodified classifier achieves, supporting the project’s plan to treat class imbalance as an explicit preprocessing concern rather than an afterthought. [4]

[5] Customer Churn Prediction in the Telecommunication Sector Using a Rough Set Approach

Feature selection is a recurring concern in churn prediction because telecom datasets often contain many correlated or redundant attributes describing usage and billing behaviour. This study applies rough set theory – a technique for identifying the minimal set of features that preserves a dataset’s classification information – to telecom churn prediction.

The rough-set approach is used to reduce the feature space before classification, aiming to retain predictive power while removing attributes that add noise rather than useful signal.

This is relevant to the proposed project’s Step 4 (Feature Selection), in which important churn-contributing factors such as tenure, recharge frequency and complaint history are expected to be identified before model training, following a similar rationale of reducing dimensionality without discarding informative variables.

The study supports the proposed system’s plan to treat feature selection as a distinct step rather than feeding all raw attributes directly into the classifier, and indicates that a carefully reduced feature set can improve both efficiency and classification quality. [5]

[6] Early Warning of Telecom Enterprise Customer Churn Based on Ensemble Learning

Early identification of at-risk customers – before they formally churn – is central to making retention strategies effective, since intervention after a customer has already left an operator is no longer possible. This study develops an early-warning framework for telecom customer churn built on ensemble-learning algorithms.

The framework analyses customer characteristics using ensemble methods to generate an early warning signal, with the explicit goal of supporting targeted marketing strategies and personalised retention services rather than simply reporting a historical churn rate.

The proposed project’s three-tier risk classification (Low/Medium/High) is conceptually aligned with this early-warning idea: rather than a binary churn/no-churn label, the system is intended to flag degrees of risk so that a telecom operator can prioritise retention effort where it is likely to matter most.

This study is relevant because it validates the practical value of framing churn prediction as an early-warning problem rather than a retrospective classification exercise, reinforcing the project’s emphasis on risk banding over a simple binary outcome. [6]

[7] Prediction of Customer Churn Behavior in the Telecommunication Industry Using Machine Learning Models

The telecommunications industry is reported to have one of the highest annual churn rates of any major industry, which motivates continued research into forecasting methods that can support timely retention intervention. This study examines ensemble-learning models for forecasting churn and incorporates explainable AI (XAI) techniques to interpret the resulting predictions.

The work combines predictive modelling with explanation methods so that the factors behind a forecasted churn risk can be identified, allowing retention teams to design focused and timely intervention strategies rather than generic, undifferentiated campaigns.

This directly supports the proposed system’s dashboard-based presentation of results: the project report plans to accompany each risk-level prediction with the leading contributing usage or billing factors, in a manner consistent with this study’s XAI-driven approach.

The study reinforces that, in a domain with structurally high churn rates such as telecommunications, prediction alone is insufficient – the explanation attached to a prediction is what ultimately enables an effective, targeted retention response. [7]

[8] Machine Learning Predictive Models Analysis on Telecommunications Service Churn Rate

Publicly available telecom churn datasets, such as the widely used Kaggle telecom-churn dataset, provide a common basis for comparing classification algorithms under consistent conditions. This study analyses and implements several machine-learning models – including XGBoost, Bernoulli Naïve Bayes and Decision Tree – to predict churn actions using such a dataset.

The comparative evaluation examines how different algorithm families perform on the same churn-labelled dataset, providing a baseline for understanding which classifiers are well suited to typical telecom subscriber attributes.

The proposed SIM-churn project follows a similar comparative philosophy at a smaller scale, planning to benchmark Logistic Regression, Decision Tree and Random Forest on a chosen telecom dataset using the same evaluation metrics (accuracy, precision, recall, F1-score, confusion matrix) so that model selection is evidence-based rather than assumed.

This study is relevant primarily as a methodological reference for structuring a comparative churn-modelling experiment, and it reinforces the value of testing more than one algorithm family before finalising a production model. [8]

[9] Machine Learning-Based Prediction of Telecom Customer Churn: Comparative Model Analysis

As churn-prediction research has matured, later studies have placed increasing emphasis on rigorous, metric-driven comparison between algorithms rather than reporting a single model's accuracy in isolation. This study conducts an empirical comparison of Logistic Regression, Random Forest and Extreme Gradient Boosting (XGBoost) for telecom churn prediction using accuracy, precision, recall, F1-score and ROC-AUC.

The evaluation finds that an optimised XGBoost model achieves the strongest performance across the tested metrics, illustrating the value of hyperparameter tuning on top of algorithm selection.

This is closely aligned with the candidate-model list in the project report (Logistic Regression, Decision Tree, Random Forest), and indicates that if the initial candidate models prove insufficient during Stage-II, boosted-tree methods such as XGBoost represent a natural extension for future work.

The study's results are specific to its own dataset and experimental setup and should not be interpreted as a guarantee of similar performance for the proposed student system; it is included because it demonstrates a rigorous, multi-metric comparison methodology that the project's

own evaluation stage is designed to follow. [9]

[10] Customer Churn Prediction in Telecom Using Machine Learning in a Big-Data Platform

Telecom operators generate subscriber data at a scale that can exceed what a single machine can process efficiently, motivating research into churn prediction on distributed, big-data platforms. This study develops a churn-prediction pipeline built on a big-data platform (Apache Spark), combining data preprocessing, feature engineering and classification at scale.

The work demonstrates that standard churn-modelling steps – cleaning, encoding and classification – can be adapted to a distributed-computing environment without changing the underlying machine-learning concepts, only the scale at which they are executed.

The proposed semester project currently operates on a single-machine, CSV/MySQL-based pipeline appropriate to its dataset size and Stage-I/Stage-II scope, but the big-data framing is relevant to the project’s future-scope direction toward handling larger, operator-scale subscriber datasets.

This study is relevant mainly as evidence that the same conceptual churn-prediction pipeline described in the project report can, in principle, scale to production-sized telecom datasets, supporting the project’s long-term extensibility rather than its immediate Stage-I implementation. [10]

[11] A Survey on Customer Churn Prediction Using Machine Learning Techniques

Given the volume of churn-prediction research, survey papers play an important role in consolidating which techniques have been applied, under what conditions, and with what general findings. This paper surveys machine-learning techniques used for customer churn prediction across multiple studies.

The survey summarises commonly used algorithms – including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, k-Nearest Neighbours and Naive Bayes – and discusses the general strengths and limitations reported for each in the churn-prediction literature.

This mirrors the project report’s own Literature Survey summary of commonly used techniques, and confirms that the algorithms selected as Stage-II candidates (Logistic Regression, Decision Tree, Random Forest) are well represented and well studied in the broader churn-prediction literature rather than being an arbitrary or unusual choice.

The survey is relevant because it establishes the wider methodological context in which the proposed churn-prediction system is being developed, supporting the project's choice of established, well-understood classification algorithms for its first modelling stage. [11]

[12] Customer Churn Prediction Using Machine Learning Models

A recurring theme in churn-prediction research is the combination of classification with an analysis of which customer factors actually drive churn, since a prediction alone does not tell a business why a customer is at risk. This study proposes a churn-prediction approach that uses both classification and clustering techniques to identify churn customers and the factors contributing to churn in the telecom sector.

The study compares XGBoost, Random Forest, k-Nearest Neighbours, Support Vector Machine and Decision Tree, reporting that XGBoost and Random Forest achieved the highest prediction accuracy among the models tested, ahead of KNN, SVM and Decision Tree, on a moderately imbalanced telecom dataset.

This is directly relevant to the proposed project's planned Feature Selection step (Step 4) and its ambition to provide *important feature analysis* alongside the churn prediction itself, rather than a risk label without supporting explanation, as noted in the project's comparative-analysis table.

The study's reported accuracy figures are specific to its own dataset and class-imbalance ratio and should not be transferred to the proposed student system; it is included because it reinforces both the competitiveness of Random-Forest-family models and the value of combining prediction with factor analysis, both of which the project report adopts as design goals. [12]

2 Research Gap

The reviewed literature shows that machine learning, and particularly ensemble and boosted-tree methods, is increasingly effective at telecom churn classification, but the focus and implementation level vary considerably across studies. Some studies emphasise raw predictive accuracy on a fixed public dataset, while others focus on explainability, class-imbalance handling, feature reduction or distributed-computing scale. The project report identifies a practical gap between these modelling studies and a lightweight, dashboard-based tool that a telecom analyst or retention-team member can use directly.

First, several research systems report high accuracy on well-known public datasets such as the Kaggle Telco Customer Churn dataset, but operator-specific subscriber behaviour, plan structures and complaint patterns can differ substantially from these reference datasets. This means

that a model trained on a public dataset should not automatically be treated as a universally valid churn-prediction tool for a different telecom operator or region.

Second, there is a difference between a churn-classification model and a complete decision-support system. A model may output a churn probability or a binary churn/no-churn label without directly providing a usable, risk-banded recommendation that a non-technical user can act on. The proposed project attempts to connect model inference with a three-level risk classification (Low/Medium/High) and a dashboard that surfaces the leading contributing factors for each prediction.

Third, class imbalance between churners and non-churners is a recurring and only partially solved challenge in the literature. Several studies apply adaptive learning, evolutionary computation or resampling strategies specifically to address this imbalance, indicating that a churn classifier trained without attention to class balance risks under-predicting the minority (churn) class – precisely the customers the system is meant to identify.

Fourth, explainability remains important for practical adoption. The reviewed literature demonstrates growing use of explainable-AI techniques and tabular-model interpretation methods, showing that retention teams need to understand *why* a customer is flagged as high-risk, not just *that* they are flagged, in order to design an appropriate retention action.

Finally, churn-prediction performance can change with dataset, operator, region and time period, so a single reported accuracy value from the literature is not sufficient evidence of real-world reliability. Therefore, the Stage-II training and evaluation planned in the project report – comparing multiple candidate models using accuracy, precision, recall, F1-score and the confusion matrix on the project’s own selected dataset – is essential before the system can be considered reliable for practical retention decisions.

3 Relevance to the Proposed System

The proposed Customer Churn Prediction System for SIM Users follows a focused decision-support workflow derived from the project report. The user-facing input consists of subscriber tenure, recharge amount, recharge frequency, call usage, internet/data usage, plan type, payment behaviour, customer complaints and general service-usage indicators. These variables form the input feature vector for the planned classification model, with categorical attributes such as plan type encoded before the complete vector is passed to the trained model after input validation.

The proposed architecture is designed to separate data preprocessing, model inference, presentation and data storage, consistent with the workflow: Customer Data → Preprocessing → Exploratory Data Analysis → Feature Selection → ML Model → Evaluation → Churn Predic-

tion → Risk Level → Dashboard, as set out in the project report’s proposed methodology.

The planned database is intended to retain customer-level query information – tenure, recharge behaviour, usage figures, plan details, complaint history and timestamp – together with the resulting churn-risk prediction, so that predictions can be reviewed over time and the retention team can track how a customer’s risk level changes across billing cycles. This structure provides traceability of previous assessments and creates a basis for later review or model retraining.

The project report specifies Python with Pandas and NumPy for data processing; Matplotlib and Seaborn for visualisation; Scikit-learn for model training; MySQL for data persistence; Flask for the backend; and HTML, CSS and JavaScript for the frontend dashboard. These technologies are presented as the planned implementation stack in the project report rather than as evidence of a completed, validated production system.

The literature therefore supports the overall design direction: apply a suitable classification method to structured, multi-attribute subscriber data; address class imbalance explicitly during preprocessing; combine the predicted risk level with an explanation of contributing factors; and present the result through an accessible, dashboard-based interface. The system’s actual model accuracy, reliability and operational usefulness for a real telecom operator remain to be established during Stage-II training, validation, integration and testing.

4 Conclusion

The reviewed research demonstrates that Artificial Intelligence and Machine Learning are increasingly important tools for telecom customer-retention decision support. Studies have applied ensemble learning, evolutionary and rough-set feature techniques, explainable-AI methods, and big-data platforms to the problem of predicting which subscribers are likely to churn. Research also shows the value of combining multiple subscriber-level inputs such as tenure, recharge behaviour, usage patterns, billing and complaint history rather than relying on any single indicator.

At the same time, the literature identifies important limitations. Model performance can depend strongly on dataset composition, operator-specific behaviour, the degree of class imbalance between churners and non-churners, and the chosen evaluation metrics. Big-data and distributed-computing approaches can help systems scale to operator-sized subscriber bases, but they introduce additional infrastructure and integration requirements. Most importantly, a high predictive score on a fixed public dataset does not by itself establish that a churn-risk classification is reliable enough to drive real retention spending for every telecom operator.

The proposed Customer Churn Prediction System for SIM Users addresses a practical part of this problem by combining subscriber usage, billing and service-interaction data with a planned

classification model, then presenting the result as a Low/Medium/High risk level through a simple, dashboard-based interface. The system also separates model inference from customer-query logging, allowing predictions to be reviewed and reassessed over time. According to the project report, at the current stage the project has completed approximately 30% of the planned work, including problem identification, requirement analysis, literature survey, dataset identification, technology-stack selection and initial data preprocessing. Stage-II will complete exploratory data analysis, feature selection, model training and evaluation, backend integration, and end-to-end testing. Therefore, the proposed system should be regarded as a planned decision-support tool whose final performance will depend on future validation.

Future Scope: real-time churn prediction; integration with live telecom billing databases; a companion mobile application; real-time customer monitoring; personalised retention offers; SMS/email risk alerts; customer segmentation; advanced machine-learning/deep-learning models; explainable AI for prediction reasons; and automated retention-recommendation generation.

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